

Generalized path-connectedness in proximity spaces

OPG-59896	L1 - Tractable	Ouvert	Topology
-----------	----------------	--------	----------

Énoncé original (anglais, reproduction mot pour mot)

Let δ be a proximity.

A set A is connected regarding δ iff $\forall X, Y \in \mathcal{P}(A) \setminus \{\emptyset\} : \left(X \cup Y = A \implies X \mathrel{\delta} Y \right)$.

Conjecture The following statements are equivalent for every endofunctor μ and a set U :

- U is connected regarding μ .
- For every $a, b \in U$ there exists a totally ordered set $P \subseteq U$ such that $\min P = a$, $\max P = b$, and for every partition $\{X, Y\}$ of P into two sets X, Y such that $\forall x \in X, y \in Y: x < y$, we have $X \mathrel{[\mu]^{\ast}} Y$.

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>